Zohaib Ali Mughal

Software Engineer: LLM agents, production TypeScript backends, Python ML

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Dart, SQL, Python
LLM and voice AI: LLM tool calling, prompt design, Retell AI, OpenAI models, ElevenLabs TTS
ML: ZenML, MLflow, scikit-learn, XGBoost, LightGBM, CatBoost, KNN imputation
AI-assisted development: Claude Code, Cursor, GitHub Copilot
Backend: Node.js, NestJS, Fastify, tRPC, REST / OpenAPI, Zod, BullMQ, NATS JetStream
Data and DevOps: PostgreSQL, Prisma, Supabase, Redis, Docker, AWS, Railway, Vercel, GitHub Actions
Frontend: Next.js, React, Tailwind CSS, TanStack Query, Radix / shadcn UI
Payments and Quality: Stripe Connect, Paymob, Airwallex, Sentry, Jest, Vitest, Playwright, Testcontainers
Relevant coursework: Introduction to Machine Learning, Introduction to Artificial Intelligence, Parallel and Distributed Computing, Cloud Computing

PROFESSIONAL EXPERIENCE

Slidey LLC, Karachi **Nov 2024 – Present**
Trainee Software Engineer (Nov 2024), Software Engineer I (Mar 2025), Software Engineer II (May 2026)

ETG Platform portal.evolutiongroup.net.au | case study **Software Engineer II**

Field-service platform for an Australian electrical, security and telecom group: admin, client, supplier and technician portals. Built to stop about $5k a week lost to unbilled supplier costs and billing scheduled rather than actual hours.

- Designed a 25-package Turborepo (Next.js frontend, Fastify API, BullMQ worker): 145 tRPC routers and 1,000+ procedures over a 212-model Prisma schema (dispatch, timesheets, assets, quoting, billing); money as BigInt cents.
- Enforcement-first decision engine: one pure, unit-testable gate checks every dispatch, closeout and invoice transition, returns structured failure codes, and logs every block and manager override (8 live checks on the dispatch calendar).
- Supplier integration layer: typed adapters for eight external systems, including three trade portals with no public API (SAP Commerce and Magento storefronts) parsed server-side, plus ABN lookup and two payment feeds.
- Scored two-hop reconciliation of bank transactions to supplier invoices; jobs cannot be invoiced until costs match.

Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, BullMQ, Redis, Better Auth, Docker.

Plexor case study **Software Engineer I**

Multi-camera computer-vision platform for retail loss prevention: detects theft, POS under-ringing, phone use and loitering across 100+ stores and routes prioritised alerts to store and district managers.

- Led the web side: the realtime video and inference pipeline on AWS video services (about 700ms inference across 37+ detection models), alert routing on a Node.js backend, and role-scoped Next.js dashboards.

Stack: Next.js, React, Node.js, AWS.

Hyss hyss.io | case study **Software Engineer I**

Live-event staffing marketplace: companies hire certified freelance crew, with competency-matched shifts, call sheets, realtime chat, invoicing and payouts. One Next.js codebase serves web, iOS and Android through Capacitor.

- Two-way Google and Apple (CalDAV) calendar sync with encrypted OAuth tokens and a retryable row-locked job queue; realtime per-event crew chat on Supabase Realtime.

Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, Supabase, Stripe Connect, Capacitor, Sentry.

Conek, for A-SAFE Digital asafe.com | case study **Software Engineer I**

Quote-to-cash platform for A-SAFE, integrated with their CRM, drawing and routing tools.

- Maintained and scaled event-driven microservices on NATS JetStream with type-safe contracts and clean-architecture layering; inter-service latency under 100ms.

Stack: TypeScript, Node.js, Fastify, NATS JetStream, Next.js, React, PostgreSQL, AWS.

Sportefy App Store | Play Store | case study **Trainee Software Engineer**

Sports booking and matchmaking app live in Pakistan: Flutter on both stores, Next.js admin panel, NestJS API.

Web admin panel and API

- Next.js admin dashboard over a NestJS API: 48 domain modules, 190+ versioned REST endpoints, additive-only /v1 contract with OpenAPI drift gates in CI; no release broke older installs.

Flutter app

- 21 feature modules, 823 passing tests, BLoC with clean architecture; offline request queue with replay, automatic token refresh, fault-tolerant parsing.

Stack: NestJS, Flutter, Next.js, Prisma, PostgreSQL, Supabase, Paymob, Railway, Vercel, GitHub Actions.

TECHNICAL PROJECTS

Voice AI Patient-Intake Agent github

- Phone agent that registers new patients over a U.S. number in English or Spanish: a Retell LLM agent with a 14-section prompt and 5 tool-calling functions, saving to Postgres only after the caller confirms a read-back.
- Fastify and Zod API, Drizzle on Postgres and a Next.js dashboard; the agent's tools and the REST API share one validation schema, webhooks are signature-checked, and 53 tests run against a real Postgres engine (PGlite). Deployed on Railway and Vercel.
- Prompt, tool schemas and agent settings live in the repo and are provisioned by an idempotent sync script; tests keep the model's tool schemas in step with the API and forbid retries on write tools so one confirmation can never create two records.

Stack: TypeScript, Retell AI, OpenAI, Fastify, Zod, Drizzle, PostgreSQL, Next.js, Vitest, Railway, Vercel.

RiskSense, Loan-Default MLOps Pipeline github **Final-Year Project with Meezan Bank Limited**

- ZenML training pipeline with MLflow experiment tracking predicting loan default on 39,503 distributor-financing records: derived default labels and repayment features, KNN imputation for missing KIBOR tenures, class weighting.
- Five interchangeable classifiers (XGBoost, LightGBM, CatBoost, Random Forest, logistic regression), each run logging accuracy, F1, ROC AUC and confusion-matrix metrics to MLflow.

Stack: Python, ZenML, MLflow, scikit-learn, XGBoost, LightGBM, CatBoost.

EDUCATION

Institute of Business Administration (IBA), Karachi **Graduated 2025**
BS Computer Science. National Talent Hunt Program Scholar (top 33 of 7,290 candidates).

ACHIEVEMENTS

- **Teaching Assistant, Mobile Application Development (IBA):** mentored 100+ students through hands-on workshops; class assessment averages rose 30%.
- **Ignite Hackathon:** 1st place for an IoT software prototype built and deployed within a 2-hour limit.
- **Trading Bazaar:** runner-up for algorithmic trading strategies.